

SAIM ZAFAR

Full-Stack Developer | Machine Learning | BSIT, Bahria University Islamabad
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PROFESSIONAL SUMMARY

Full-Stack Developer and machine learning practitioner with a strong academic record (CGPA 3.90) and hands-on experience building and deploying end-to-end web and ML applications. Skilled in React, Node.js/Express, FastAPI, REST APIs, MongoDB, and MySQL, with secure authentication (JWT, OAuth) and cloud deployment via Docker, Vercel, Render, and Hugging Face Spaces. Focused on building secure, scalable, and explainable systems.

EXPERIENCE

Software Engineering Intern (MERN Stack) - 10Pearls, Shine Internship Program

Remote | Jul 2026 - Present

- Selected for 10Pearls' competitive 8-week virtual internship program (Cohort 9) focused on full-stack development with the MERN stack (MongoDB, Express.js, React, Node.js)
- Building and shipping web application features through structured sprints, code reviews, and technical assessments under mentorship from 10Pearls engineers

EDUCATION

Bachelor of Science in Information Technology (BSIT)

Bahria University, Islamabad | CGPA: 3.90 / 4.00 | Merit scholarship every eligible semester (GPAs 4.0, 3.91, 3.94)

A-Levels, Computer Science - Asas International School, Islamabad | Aug 2021 - Jan 2023 (Mathematics A, Physics B, Computer Science C)

IELTS Academic - Overall Band 7.0 (May 2023)

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, C, SQL

Frontend: React, HTML5, CSS3, JavaScript, jQuery, AJAX

Backend: Node.js, Express.js, FastAPI, REST API Design, JWT, OAuth, Role-Based Access Control

Databases: MongoDB, MySQL, Oracle

Machine Learning / Data: PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, CNNs, Grad-CAM

DevOps & Cloud: Docker, Docker Compose, Git/GitHub, Vercel, Render, Railway, Hugging Face Spaces, Supabase

Other: Blockchain (PBFT Consensus, Digital Signatures), Jest, Qt

PROJECTS

ECG-Scope - Explainable ECG Diagnosis Platform (Deep Learning, Full-Stack) [Live Demo](#) | [GitHub](#)

Tech: Python, PyTorch, FastAPI, React, Supabase, Docker, scikit-learn, Hugging Face Spaces, Vercel

- Trained a 1D CNN from scratch on the PTB-XL dataset (~21,000 ECGs) to classify 12-lead ECGs into 5 diagnostic categories, achieving 0.92 macro-AUC with patient-disjoint splits and Grad-CAM visual explanations
- Diagnosed BatchNorm covariate shift in a 3-hospital cross-dataset study and applied label-free test-time adaptation (AdaBN) to recover 0.87 macro-AUC
- Deployed a Dockerized FastAPI model API on Hugging Face Spaces with a React frontend on Vercel, Supabase authentication, and per-user analysis history

Truvex - Permissioned Blockchain for Academic Credential Verification [GitHub](#)

Tech: Node.js, Express.js, React, MongoDB Atlas, Docker, JWT, PBFT Consensus, Digital Signatures

- Engineered a permissioned blockchain running PBFT consensus across 7 validator nodes with view-change leader failure recovery and cryptographically signed transactions
- Built REST APIs for credential issuance, verification, and revocation with schema validation and per-node persistent ledgers; validated fault tolerance through adversarial testing (forged signatures, malicious leader)
- Containerized validators with Docker Compose, deployed nodes to Railway, and implemented JWT authentication with role-based React dashboards for regulator, university, and employer roles

ModGuard - AI Content Moderation Platform (Full-Stack, AI Integration) [Live Demo](#) | [GitHub](#)

Tech: Node.js, Express.js, MongoDB, React, JWT, Google Gemini API, Docker

- Built a full-stack platform screening user-submitted images via the Google Gemini API across 6 policy categories, returning Approved/Flagged/Blocked verdicts with per-category confidence and reasoning

- Designed a policy-driven verdict engine with admin-configurable thresholds and per-submission policy snapshots for a reproducible audit trail
- Implemented JWT authentication with role-based access control, a user appeals workflow, a Recharts analytics dashboard, and Docker Compose containerization

Nexora - Full-Stack Online Marketplace (Team Project) [GitHub](#)

Tech: Node.js, Express.js, MySQL, Sequelize, AWS S3, JWT, Passport.js, Vercel, Render

- Architected RESTful APIs for user management, product listings, cart, checkout, and order history; secured with bcrypt password hashing and Helmet
- Implemented multi-strategy authentication (Google OAuth, Facebook OAuth, local login) using Passport.js and JWT; integrated AWS S3 for scalable image storage
- Deployed a cloud-native architecture (Vercel frontend, Render backend, Clever Cloud MySQL) with an admin panel, CSV export, and QR-code order generation

SkillLink - Freelance Marketplace System [GitHub](#)

Tech: Oracle XE 21c, Node.js, Express.js, React.js, REST API, JWT, Role-Based Access Control

- Built a full-stack freelance marketplace on 15 fully normalized (3NF) Oracle tables covering registration, project posting, competitive bidding, contracts, milestones, and reviews
- Exposed 35 REST API endpoints with multi-table transactions and role-based access control for clients, freelancers, and admins
- Runner-Up, IEEE Department Innovation Challenge - Semester Project Showcase (Spring 2026), Bahria University

NovaCORE - OS Concepts Simulator with xv6 Kernel Extension [GitHub](#)

Tech: C (C99), POSIX pthreads, GCC, QEMU, xv6 Kernel

- Built a terminal-based OS simulator covering CPU scheduling (Round Robin/SJF/Priority), page replacement (FIFO/LRU/Optimal), Producer-Consumer IPC, and deadlock detection (Banker's Algorithm)
- Extended the xv6 teaching kernel with a custom getmeminfo() system call across 9 kernel source files for live physical-memory reporting inside QEMU; compiled with zero warnings (GCC 13, -Wall)

Competria - Competitive Programming Platform [GitHub](#)

Tech: C++, Qt Framework, QScintilla, OOP

- Built a desktop problem-solving platform with role-based access control, secure authentication, dynamic problem creation, and automated code evaluation with real-time feedback
- Integrated the QScintilla code editor for in-app coding within a modular OOP architecture

Python Data Science & ML Implementations [GitHub](#)

Tech: Python, NumPy, Pandas, scikit-learn, Matplotlib

- Implemented KNN, Decision Trees (ID3), Random Forest, SVM, and PCA from scratch and benchmarked against scikit-learn across an end-to-end workflow (preprocessing, training, hyperparameter tuning, evaluation)

ACHIEVEMENTS & CERTIFICATIONS

- Merit-Based Scholarship (1st Position) - awarded in the 1st, 2nd, and 4th semesters at Bahria University
- Rector's Honor List - 1st, 2nd, and 4th semesters
- 5th Place - CODEZAAR Competitive Programming Competition, Bahria University (multi-university event)
- Runner-Up (IEEE Award) - Department Innovation Challenge, Semester Project Showcase, Bahria University (Spring 2026)
- Python for Data Science - Udemy
- Participant - Youth Speak Forum 2025 (AIESEC Islamabad)